RAG Quick Guide (sample document for the RAG Chatbot project)

What is RAG? Retrieval-Augmented Generation (RAG) is a way to make a language model answer questions using your own documents. First the system retrieves the passages that are most related to the question. Then a language model generates an answer using only those passages.

Why use RAG? A language model only knows what was in its training data, and it can invent confident but wrong answers. This is called hallucination. RAG reduces this problem because the model is given real text to quote, and the app can show the sources it used.

Step 1, chunking. Long documents are cut into small overlapping pieces called chunks. In this project a chunk is about 500 characters long, and the last few words of each chunk are repeated at the start of the next one so that a sentence is not lost at a boundary.

Step 2, embeddings. Each chunk is turned into an embedding: a list of 384 numbers produced by the all-MiniLM-L6-v2 model. Texts with similar meaning get similar numbers, even when they use different words.

How the chatbot answers a question

Step 3, the vector database. The embeddings are stored in ChromaDB, a vector database. When a question arrives, it is embedded with the same model, and ChromaDB returns the four chunks whose embeddings are closest to it. Closeness is measured with cosine distance: a smaller distance means a closer meaning.

Step 4, generation. The retrieved chunks, the last few chat messages and the question are sent to a language model with the instruction to answer only from the given context. If no language model is available, the app shows the best matching passages instead.

Good questions to try: What is hallucination? How long is a chunk? Which model creates the embeddings? What does a smaller cosine distance mean?

Tip: to add your own documents, copy PDF files into the docs folder and run python ingest.py, or use the Upload PDF button in the web page. Scanned PDFs that contain only pictures of text cannot be read, because they have no text layer.